[BUG] op_math adding two GellMann observables doesn't treat them as di

https://github.com/PennyLaneAI/pennylane/issues/4341

ROW 30

[BUG] op_math adding two GellMann observables doesn't treat them as different Hermitians #4341

New issue

Closed | #4366

Gabriel-Bottrill opened on Jul 7, 2023

Expected behavior

```
>>> gell_1 = qml.GellMann(wires=0, index=3)
>>> gell_2 = qml.GellMann(wires=0, index=8)
>>> ham_added = gell_1 + gell_2
>>> print(type(ham_added))
<class 'pennylane.ops.qubit.hamiltonian.Hamiltonian'>
>>> print(ham_added.coeffs)
array([1,1])
>>> print(ham_added.ops)
[GellMann3(wires=[0]), GellMann8(wires=[0])]
```

The two GellMann observables should be appended in a Hamiltonian object as shown.

Actual behavior

```
>>> gell_1 = qml.GellMann(wires=0, index=3)
>>> gell_2 = qml.GellMann(wires=0, index=8)
>>> ham_added = gell_1 + gell_2
>>> print(type(ham_added))
<class 'pennylane.ops.qubit.hamiltonian.Hamiltonian'>
>>> print(ham_added.coeffs)
array([2])
>>> print(ham_added.ops)
[GellMann3(wires=[0])]
```

Currently when adding two GellMann observables Pennylane interprets them as the same type and concatinates them together taking the first as it's type.

Additional information

By using `qml.operation.enable_new_opmath()` the bahaviour seems to work correctly.

```
>>> qml.operation.enable_new_opmath()

>>> gell_1 = qml.GellMann(wires=0, index=3)
>>> gell_2 = qml.GellMann(wires=0, index=8)
>>> ham_added = gell_1 + gell_2
>>> print(type(ham_added))
<class 'pennylane.ops.op_math.sum.Sum'>
>>> print(ham_added)
GellMann3(wires=[0]) + GellMann8(wires=[0])
```

Similar to #3114, on comment chain I was told to make a new issue

Source code

No response

Tracebacks

No response

System information

Assignees

mudit2812

Labels

bug 🐛

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

Fix `qml.GellMann` observable data
PennyLaneAI/pennylane

Notifications Customize

Subscribe

You're not receiving notifications from this thread.

Participants

Tracebacks

No response

System information

```
Name: PennyLane
Version: 0.32.0.dev0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: https://github.com/PennyLaneAI/pennylane
Author:
Author-email:
License: Apache License 2.0
Location: /usr/local/lib/python3.10/dist-packages
Requires: appdirs, autograd, autoray, cachetools, networkx, numpy, pennylane-lightning, requests, rustworkx, s
Required-by: PennyLane-Lightning

Platform info:           Linux-5.15.107+-x86_64-with-glibc2.31
Python version:          3.10.12
Numpy version:           1.22.4
Scipy version:           1.10.0
Installed devices:
- default.gaussian (PennyLane-0.32.0.dev0)
- default.mixed (PennyLane-0.32.0.dev0)
- default.qubit (PennyLane-0.32.0.dev0)
- default.qubit.autograd (PennyLane-0.32.0.dev0)
- default.qubit.jax (PennyLane-0.32.0.dev0)
- default.qubit.tf (PennyLane-0.32.0.dev0)
- default.qubit.torch (PennyLane-0.32.0.dev0)
- default.qutrit (PennyLane-0.32.0.dev0)
- null.qubit (PennyLane-0.32.0.dev0)
- lightning.qubit (PennyLane-Lightning-0.31.0)
```

Existing GitHub issues

- [x] I have searched existing GitHub issues to make sure the issue does not already exist.

Gabriel-Bottrill added bug on Jul 7, 2023

albi3ro on Jul 10, 2023 Contributor

Thanks for opening this issue @Gabriel-Bottrill .

We could potentially override the dunder methods on qutrit observables so they always use operator arithmetic.

timmysilv assigned mudit2812 on Jul 13, 2023

mudit2812 mentioned this on Jul 18, 2023

Fix `qml.GellMann` observable data #4366

mudit2812 closed this as completed in #4366 on Jul 18, 2023

Fix `qml.GellMann` observable data #4366

Code

Merged mudit2812 merged 4 commits into `master` from `gm-hamiltonian` on Jul 18, 2023

Conversation 9 | Commits 4 | Checks 0 | Files changed 5 | +60 −0

mudit2812 commented on Jul 18, 2023 • edited · Contributor

Context:
Fixes #4341 .

Description of the Change:

- Update `Observable._obs_data()` to include the index of `qml.GellMann` in the parameter list.
- Update `Hamiltonian._obs_data()` to include the index of `qml.GellMann` in the parameter list.

Note:
I would still recommend using the new operator arithmetic with qutrits. The generators of parametric qutrit gates are instances of `Prod` and `Sum`, not `qml.Hamiltonian`. So, try using `qml.dot` instead of `qml.Hamiltonian` with qutrit observables.

Benefits:
Comparison of instances of `qml.GellMann`, `Hamiltonian`s and `Tensor`s containing `qml.GellMann` is correct.

Possible Drawbacks:

Related GitHub Issues:
#4341

`Fixed bug` 1ae1ec8

mudit2812 requested a review from **a team** 2 years ago

mudit2812 commented on Jul 18, 2023 · Contributor · Author

[sc-41597]

`Merge branch 'master' into gm-hamiltonian` Verified 3cc8129

mudit2812 commented on Jul 18, 2023 · View reviewed changes

`doc/releases/changelog-dev.md` Outdated · Show resolved

`Update doc/releases/changelog-dev.md` Verified d388605

Reviewers
timmysilv ✓
albi3ro ✓

Assignees
No one assigned

Labels
None yet

Projects
None yet

Milestone
No milestone

Development
Successfully merging this pull request may close these issues.
[BUG] op_math adding two GellMann observ...

Notifications · Customize
Subscribe
You're not receiving notifications from this thread.

3 participants